WORKED SOLUTION

STEP I 2013, Question 11

Mechanics • Friction and Equilibrium

STEP I 2013, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q11

The key idea

The two strings make angles $\frac{1}{2}\pi - \alpha$ and α with the horizontal on opposite sides, so they are *perpendicular*. Two perpendicular forces $T\cos\beta$ and $T\sin\beta$ combine into a single force of size T, turned through β from the first one. Once the two strings are replaced by one pull of T at angle $\alpha + \beta$ to the horizontal, both parts become short exercises: limiting friction in (i), and in (ii) the realisation that the block is lifted clean off the surface.

Solution

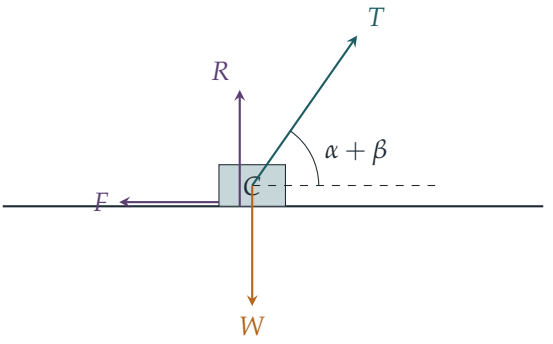
We model C as a particle and the strings as light. Take the x-axis horizontal towards B and the y-axis vertically up.

The resultant of the tensions

The string BC points at angle α above the positive x-axis, so its unit vector is $(\cos\alpha, \sin\alpha)$. The string AC makes angle $\frac{1}{2}\pi - \alpha$ with the negative x-axis, so it points at angle $\pi - (\frac{1}{2}\pi - \alpha) = \frac{1}{2}\pi + \alpha$ to the positive x-axis, with unit vector $(-\sin\alpha, \cos\alpha)$. These are perpendicular. The total pull of the strings is therefore

$$\begin{aligned} T\cos\beta\,(\cos\alpha, \sin\alpha) + T\sin\beta\,(-\sin\alpha, \cos\alpha) &= T(\cos\alpha\cos\beta - \sin\alpha\sin\beta, \\ &\qquad \sin\alpha\cos\beta + \cos\alpha\sin\beta) \\ &= T(\cos(\alpha+\beta),\ \sin(\alpha+\beta)). \end{aligned}$$

So the strings act exactly like a single force of magnitude T at angle $\alpha + \beta$ above the horizontal, pulling towards the B side. Since $0 < \alpha + \beta < \frac{1}{2}\pi$, both components are positive.



Part (i)

Let R be the normal reaction and F the friction force, which (if the block is at rest) acts horizontally away from B. For equilibrium,

$$\begin{aligned} \text{vertically:} \quad & R + T\sin(\alpha+\beta) = W, \qquad && \text{so } R = W - T\sin(\alpha+\beta), \\ \text{horizontally:} \quad & F = T\cos(\alpha+\beta). \end{aligned}$$

The condition $W > T\sin(\alpha+\beta)$ makes $R > 0$, so the block really is pressed onto the surface and friction is available. The block stays at rest provided the friction needed does not exceed its limiting value, $F \le \mu R$:

$$T\cos(\alpha+\beta) \le \mu(W - T\sin(\alpha+\beta)).$$

Now put $\mu = \tan\lambda = \sin\lambda/\cos\lambda$ and multiply through by $\cos\lambda$, which is positive because λ is acute:

$$T\cos(\alpha+\beta)\cos\lambda \le W\sin\lambda - T\sin(\alpha+\beta)\sin\lambda.$$

Rearranging and using $\cos(P-Q) = \cos P\cos Q + \sin P\sin Q$,

$$W\sin\lambda \ge T(\cos(\alpha+\beta)\cos\lambda + \sin(\alpha+\beta)\sin\lambda) = T\cos(\alpha+\beta-\lambda),$$

which is the required condition. (Every step is reversible, so in fact the block stays at rest *if and only if* this holds.)

? **Ask yourself** Why do we need the condition $W > T\sin(\alpha+\beta)$ at all?

Without it, the vertical equation would give $R \le 0$. A surface can push but not pull, so a negative R is impossible: the block would be lifted off (or only just touching), and there would be no friction to hold it. That is exactly the situation of part (ii).

Part (ii)

Now $W = T\tan\phi$ with $\alpha + \beta = 2\phi$, so $0 < \phi < \frac{1}{4}\pi$. Compare the weight with the upward pull of the strings:

$$\begin{aligned} T\sin(\alpha+\beta) - W &= T(2\sin\phi\cos\phi - \tan\phi) \\ &= T\tan\phi\,(2\cos^2\phi - 1) \\ &= T\tan\phi\cos 2\phi > 0, \end{aligned}$$

since $0 < \phi < \frac{1}{4}\pi$ makes both $\tan\phi$ and $\cos 2\phi$ positive. So the strings pull upwards harder than gravity pulls down. If the block stayed on the surface we would need $R = W - T\sin(\alpha+\beta) < 0$, which is impossible. Hence the block leaves the surface at once: $R = 0$, and with no contact there is no friction either.

The only forces are then the pull T at angle 2ϕ and the weight. The resultant force is

$$(T\cos 2\phi,\ T\sin 2\phi - T\tan\phi) = (T\cos 2\phi,\ T\tan\phi\cos 2\phi) = \frac{T\cos 2\phi}{\cos\phi}\,(\cos\phi,\ \sin\phi),$$

using the computation above for the vertical component. Its magnitude $T\cos 2\phi/\cos\phi$ is positive, so it is not zero. The block starts from rest, so its initial acceleration, and hence the direction in which it starts to move, is along the resultant force. That direction is $(\cos\phi, \sin\phi)$, at angle ϕ above the horizontal (towards B), as required.

Checks

- Take $T = 10\,\text{N}$, $\alpha = 20°$, $\beta = 40°$, so $\phi = 30°$ and $W = 10\tan 30° \approx 5.77\,\text{N}$. The upward pull is $10\sin 60° \approx 8.66\,\text{N} > W$, and the resultant is $(10\cos 60°,\ 8.66 - 5.77) = (5,\ 2.89)$, whose angle is $\arctan(2.89/5) = 30° = \phi$.
- In (i), $\lambda = \frac{1}{2}\pi$ (infinitely rough) gives $W \geq T\sin(\alpha+\beta)$, which already holds, so the block never slides; $\lambda \to 0$ (smooth) gives $0 \geq T\cos(\alpha+\beta)$, which fails, so a smooth surface cannot hold it. Both are what we expect.
- The condition in (i) was also confirmed by computer against the direct test $F \leq \mu R$ on many random choices of α, β, T, W, μ.

Common slips

- Resolving each string separately and getting lost in the algebra, instead of noticing that the strings are perpendicular and combining them first.
- Writing $R = W - T\sin(\alpha+\beta)$ in part (ii) and carrying on with friction, without noticing that R would be negative.
- Forgetting to justify that a particle starting from rest moves initially along the resultant force.

Beyond STEP

The idea of λ, the *angle of friction*, is that the normal reaction and limiting friction combine into a single total reaction making angle λ with the normal; equilibrium is possible exactly when the total reaction can lie inside this "cone of friction". Part (i) is that statement in disguise, and the same picture handles friction in three dimensions and in rigid-body problems. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction, constraints and contact forces carefully.

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